Complex Differential Geometry on the Helical Manifold

Parts I–II: Single Cell & Stacked Chain — Axioms 1–6, Corollaries 1–10

Contraction Integrals Throughout · Leibniz Notation Only

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March 2026

Provisional Draft — Not Yet Submitted for Peer Review

Prerequisite Reading

This paper uses contraction integrals exclusively—no Riemann sums appear anywhere. The contraction integral is an alternative geometric framework for integration based on level-set contraction and the layer-cake decomposition. Its central tool is **inverse duality**: for a strictly monotone continuous f on $[a, b]$,

$$\int_a^b f + \int_{f(a)}^{f(b)} f^{-1} = b \cdot f(b) - a \cdot f(a).$$

The companion paper, "The Contraction Integral: Integration via Level-Set Contraction" (Yazdani, 2026), develops this framework from first principles and should be read first.

Abstract

Six axioms define a helical 2-manifold embedded as a surface in $\mathbb{R}^3$. From these axioms, together with standard differential geometry (explicitly imported where required), we obtain: an induced metric with $g^{uu} = \frac{1}{2}$ (fixing $c = 1$); a constant gauge connection $A_u = 2\pi k$ whose holonomy identifies k as a topological winding number; a gauge-diagonalised cone metric; geodesic-derived radial dynamics with an effective potential $V(u) = k^2/(2(1+u)^2)$; a proof that local physics is cell-independent; integer-quantised angular momentum from the closure condition; a per-mode radial ODE (Euler's equation on the cone); and a phase-matching condition $\ell = k$ from holonomy coherence.

The radial ODE admits: interpretation as the separated form of a wave equation on the manifold; a scalar first-order factorisation analogous to relativistic factorisation procedures; and reduction to a Schrödinger-type normal form.

The Laplace–Beltrami operator and wave equation $\Box_{\mathcal{M}}\psi = 0$ are **imported from standard Riemannian geometry** and are not derived from the axioms. Any physical interpretation (Klein–Gordon, Dirac, Schrödinger analogies) is presented as **structural correspondence**, not equivalence.

All integrals are contraction integrals computed via inverse duality. All derivatives in Leibniz notation. One manifold. One parameter.

1 The Six Axioms

Axiom 1 (Time is Self-Referential)**.**

$$\frac{dt}{dt} = 1.$$

This fixes $c = 1$ in natural units. The z-axis grows as $z = 1 + u$, with $u \in [0, 1]$ the affine parameter within one cell.

Axiom 2 (Coupled Radius)**.** *Radius is derived from the angular variable ($r = \theta/\pi$) and the spiral expands as a light cone: $dr/dz = 1$. For one cell:*

$$r(u) = 1 + u.$$

Axiom 3 (Light-Cone Expansion)**.** *The radial expansion rate equals the speed of light: $dr/dz = 1$. Combined with Axiom 2, this couples the transverse radius to the longitudinal coordinate.*

Axiom 4 (Helical Motion)**.**

$$\theta(u, v) = v + 2\pi k\, u, \qquad k \in \mathbb{Z}^+.$$

The winding number k is the single free geometric parameter.

Axiom 5 (Reality Embedding)**.**

$$\mathbf{R}(u, v) = \big((1 + u)\cos\theta,\ (1 + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k\, u.$$

Axiom 6 (Unitary Closure)**.** *The spiral closes at integer z:*

$$\psi(u, v + 2\pi) = \psi(u, v).$$

This forces every transverse mode to carry integer angular momentum.

2 Corollaries

Corollary 1 (Induced Metric)**.** *The first fundamental form on $\mathcal{M}$ is computed by taking partial derivatives of the embedding $\mathbf{R}(u, v)$ with respect to u and v, then forming dot products.*

The tangent vectors are:

$$\mathbf{R}_u = \frac{\partial \mathbf{R}}{\partial u}, \qquad \mathbf{R}_v = \frac{\partial \mathbf{R}}{\partial v}.$$

The metric components $E = \mathbf{R}_u \cdot \mathbf{R}_u$, $F = \mathbf{R}_u \cdot \mathbf{R}_v$, $G = \mathbf{R}_v \cdot \mathbf{R}_v$ evaluate to:

$$E = 4\pi^2 k^2 (1 + u)^2 + 2, \tag{1}$$

$$F = 2\pi k\,(1 + u)^2, \tag{2}$$

$$G = (1 + u)^2. \tag{3}$$

The metric determinant and its square root:

$$\det(g) = EG - F^2 = 2(1 + u)^2, \qquad \sqrt{g} = \sqrt{2}\,(1 + u).$$

The inverse metric component:

$$g^{uu} = \frac{G}{\det(g)} = \frac{(1 + u)^2}{2(1 + u)^2} = \frac{1}{2}.$$

Derivation. *The embedding gives:*

$$\mathbf{R}_u = \big(\cos\theta - (1+u)(2\pi k)\sin\theta,\ \ \sin\theta + (1+u)(2\pi k)\cos\theta,\ \ 1\big),$$
$$\mathbf{R}_v = \big(-(1+u)\sin\theta,\ \ (1+u)\cos\theta,\ \ 0\big).$$

Computing dot products and applying $\sin^2\theta + \cos^2\theta = 1$*:*

$$\begin{aligned} E &= \cos^2\theta + (1+u)^2(2\pi k)^2\sin^2\theta - 2(1+u)(2\pi k)\cos\theta\sin\theta \\ &\quad + \sin^2\theta + (1+u)^2(2\pi k)^2\cos^2\theta + 2(1+u)(2\pi k)\sin\theta\cos\theta + 1 \\ &= 1 + 4\pi^2k^2(1+u)^2 + 1 = 4\pi^2k^2(1+u)^2 + 2. \end{aligned}$$

The cross terms in $\sin\theta\cos\theta$ *cancel. Similarly, the cross terms cancel in* F *and* G*, yielding the stated results. The determinant follows from direct computation:*

$$EG - F^2 = \big(4\pi^2k^2(1+u)^2 + 2\big)(1+u)^2 - 4\pi^2k^2(1+u)^4 = 2(1+u)^2.$$

Verified symbolically via SymPy.

Corollary 2 (Holonomy — Winding Number as Topological Charge)**.** *The off-diagonal metric component divided by the transverse metric component defines a gauge connection for the* v*-fibre:*

$$A_u = \frac{F}{G} = \frac{2\pi k\,(1+u)^2}{(1+u)^2} = 2\pi k.$$

Interpretation. A_u *measures how much phase (the* v*-direction) gets dragged per step in* u*. Because both* F *and* G *scale as* $(1+u)^2$*, their ratio is constant: the helix winds at a uniform rate regardless of where you are in the cell.*

The holonomy over one cell:

$$\Phi_{\text{cell}} = \int_0^1 A_u\,du = \int_0^1 2\pi k\,du = 2\pi k.$$

This is exact: k *complete phase cycles per cell. Since* $k \in \mathbb{Z}^+$*, the phase returns to itself* ($e^{i\cdot 2\pi k} = 1$)*, making* k *a* ***topological invariant****—it cannot be changed by continuous deformation of the manifold.*

The holonomy selects k *as the fundamental geometric quantum number of the helix.*

Verified symbolically via SymPy.

Corollary 3 (Christoffel Symbols)**.** *Computed from the standard formula*

$$\Gamma^i_{jl} = \frac{1}{2}\sum_m g^{im}\left(\frac{\partial g_{mj}}{\partial x^l} + \frac{\partial g_{ml}}{\partial x^j} - \frac{\partial g_{jl}}{\partial x^m}\right),$$

using the metric (1)–(3) *and its inverse. The six independent symbols are:*

$$\Gamma^u_{uu} = -2\pi^2k^2(u+1), \qquad \Gamma^v_{uu} = \frac{4\pi k\big(\pi^2k^2u^2 + 2\pi^2k^2u + \pi^2k^2 + 1\big)}{u+1}, \tag{4}$$

$$\Gamma^u_{uv} = -\pi k\,(u+1), \qquad \Gamma^v_{uv} = \frac{2\pi^2k^2u^2 + 4\pi^2k^2u + 2\pi^2k^2 + 1}{u+1}, \tag{5}$$

$$\Gamma^u_{vv} = -\frac{u+1}{2}, \qquad \Gamma^v_{vv} = \pi k\,(u+1). \tag{6}$$

Verified symbolically via SymPy and cross-checked numerically at $u = 1/2$***,*** $k = 1$***.***

Corollary 4 (Cell Action via Contraction)**.** *The action (surface area) of one cell:*

$$S = \int_0^{2\pi} \int_0^1 \sqrt{g}\, du\, dv = 2\pi \int_0^1 \sqrt{2}\,(1+u)\, du.$$

Set $h(u) = \sqrt{2}\,(1+u)$. Then $h(0) = \sqrt{2}$, $h(1) = 2\sqrt{2}$, and $h^{-1}(t) = t/\sqrt{2} - 1$.
By inverse duality:

$$\int_0^1 h(u)\, du = 1 \cdot h(1) - 0 \cdot h(0) - \int_{\sqrt{2}}^{2\sqrt{2}} h^{-1}(t)\, dt = 2\sqrt{2} - \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}.$$

Therefore:

$$S = 2\pi \times \frac{3\sqrt{2}}{2} = 3\sqrt{2}\,\pi.$$

The action is ***independent of*** *k. The winding number does not affect the surface area of the cell.*

Verified symbolically via SymPy.

Corollary 5 (Phase Energy via Geodesic Derivation)**.** ***Origin of the potential.*** *The coordinate change $v' = v + 2\pi k u$ removes the off-diagonal metric term, yielding the cone metric:*

$$ds^2 = 2\, du^2 + (1+u)^2\, dv'^2.$$

The geodesic equations on this cone are:

$$\frac{d^2u}{ds^2} - \frac{(1+u)}{2}\left(\frac{dv'}{ds}\right)^2 = 0,$$
$$\frac{d}{ds}\left[(1+u)^2 \frac{dv'}{ds}\right] = 0.$$

The second equation gives conservation of angular momentum: $(1+u)^2\, dv'/ds = L = const.
Substituting $dv'/ds = L/(1+u)^2$ into the first equation:

$$\frac{d^2u}{ds^2} = \frac{L^2}{2(1+u)^3}.$$

Reducing to first order via $p\, dp/du = d^2u/ds^2$ and integrating:

$$p^2 = E_{total} - \frac{L^2}{2(1+u)^2}.$$

From the metric normalisation $2p^2 + L^2/(1+u)^2 = 1$, the total energy is $E_{total} = 1/2 = g^{uu}$.
The position-dependent term is the geometric potential:

$$V(u) = \frac{k^2}{2(1+u)^2},$$

where we identify $L = k$ (the angular momentum equals the winding number).

Integration. *The integrated potential per cell:*

$$E_{phase} = \int_0^1 \frac{k^2}{2(1+u)^2}\, du.$$

By the layer-cake decomposition, the level width of $V(u) = k^2/(2(1+u)^2)$ (strictly decreasing on $[0,1]$) is:

$$w_V(t) = \begin{cases} 1 & 0 \le t < k^2/8, \\ \dfrac{k}{\sqrt{2t}} - 1 & k^2/8 \le t \le k^2/2. \end{cases}$$

Contraction integral:

$$\int_0^{k^2/2} w_V(t)\,dt = \frac{k^2}{8} + \frac{k^2}{8} = \frac{k^2}{4}.$$

$$\boxed{E_{phase} = \frac{k^2}{4}.}$$

Consistency check: *Integrated kinetic energy + integrated potential* $= (1/2 - k^2/4) + k^2/4 = 1/2 = g^{uu}$. ✓

Verified symbolically via SymPy.

Corollary 6 (Cell-Independence of the Local Metric)**.** ***Statement.*** *The quantities governing local physics on each cell are independent of the cell number n.*

Proof. *Parameterise cell n with base radius a (where $a = n + 1$ for cell n in the global chain), so the embedding becomes:*

$$\mathbf{R}_a(u, v) = \big((a + u)\cos\theta,\ (a + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k u.$$

The metric components on this general cell are:

$$\begin{aligned} E(a) &= 4\pi^2 k^2 (a + u)^2 + 2, \\ F(a) &= 2\pi k\,(a + u)^2, \\ G(a) &= (a + u)^2. \end{aligned}$$

These do *depend on a. However, the physically governing quantities do not:*

1. Gauge connection:

$$A_u = \frac{F(a)}{G(a)} = \frac{2\pi k\,(a + u)^2}{(a + u)^2} = 2\pi k. \qquad \textit{(independent of } a\textit{)}$$

2. Inverse metric:

$$g^{uu} = \frac{G(a)}{\det g(a)} = \frac{(a + u)^2}{2(a + u)^2} = \frac{1}{2}. \qquad \textit{(independent of } a\textit{)}$$

3. Gauge-diagonalised metric: *The coordinate change $v' = v + 2\pi k u$ gives*

$$ds^2 = 2\,du^2 + (a + u)^2\,dv'^2$$

on every cell. The wave equation $\Box\psi = 0$ with $\psi = F(u)\,e^{i\ell v'}$ yields Euler's ODE:

$$(a + u)^2 F'' + (a + u)F' - 2\ell^2 F = 0,$$

whose indicial roots $r = \pm\ell\sqrt{2}$ are ***independent of*** *a.*

What does depend on *a:* *The ratio $r(1)/r(0) = (a + 1)/a$, which equals 2 only when $a = 1$. The transfer matrix trace $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$ uses $\ln(w(1)/w(0)) = \ln 2$, which requires $a = 1$.*

Conclusion. *The chain is periodic: each cell resets to $a = 1$, with $u \in [0, 1]$ as a local coordinate. The gauge connection, the inverse metric, and the indicial structure of the wave equation are cell-independent. The transfer matrix (derived in later corollaries) requires this periodicity to achieve $\det T = 1/2$ and $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$.*

Verified symbolically via SymPy.

3 Part II: The Stacked Chain

Corollary 7 (Fourier Decomposition on the Transverse Circle)**.** *Axiom 6 requires $\psi(u, v+2\pi) = \psi(u, v)$. A function $e^{i\ell v}$ satisfies this if and only if:*

$$e^{i\ell(v+2\pi)} = e^{i\ell v} \quad \Longleftrightarrow \quad e^{2\pi i\ell} = 1 \quad \Longleftrightarrow \quad \ell \in \mathbb{Z}.$$

Non-integer values fail: for example, $\ell = 1/2$ gives $e^{i\pi} = -1$, which would require $\psi(u, v + 2\pi) = -\psi(u, v)$—a weaker closure condition not permitted by Axiom 6.

The general solution on cell n is therefore:

$$\psi_n(u, v) = \sum_{\ell=-\infty}^{\infty} c_\ell^{(n)} f_\ell^{(n)}(u)\, e^{i\ell v}, \qquad \ell \in \mathbb{Z}.$$

No truncation. The full tower of integer angular momentum modes is present. The quantisation of ℓ is not imposed—it is forced by the closure condition.

Verified symbolically via SymPy *(integer ℓ gives $e^{2\pi i\ell} = 1$; half-integer gives -1).*

Corollary 8 (Per-Mode Radial ODE)**.** ***Imported machinery.*** *The Laplace–Beltrami operator is the unique second-order differential operator on a Riemannian manifold that is (a) coordinate-invariant, (b) self-adjoint, and (c) reduces to the ordinary Laplacian on flat space. It is* not derived from the axioms. *We apply known Riemannian geometry to the metric our axioms produced:*

$$\Box_{\mathcal{M}}\psi = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i}\left(\sqrt{g}\, g^{ij} \frac{\partial \psi}{\partial x^j}\right).$$

Two correction factors versus the flat Laplacian: g^{ij} accounts for non-orthogonal coordinate directions; $\sqrt{g}$ accounts for varying patch area across the surface.

In the original *(u, v)* ***frame.*** *Substituting $\psi = f(u)\, e^{i\ell v}$ into $\Box_{\mathcal{M}}\psi = 0$ and dividing by $e^{i\ell v}$ yields a second-order ODE for $f(u)$ that depends on both k (from the metric) and ℓ (from the Fourier mode). The full expression is large due to the off-diagonal metric coupling.*

In the gauge-diagonalised frame. *The coordinate change $v' = v+2\pi ku$ reduces the metric to the cone $ds^2 = 2\, du^2 + (1+u)^2\, dv'^2$. The same Laplace–Beltrami operator with $\psi = F(u)\, e^{i\ell v'}$ yields Euler's ODE:*

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0.$$

The indicial equation $r^2 = 2\ell^2$ gives roots $r = \pm\ell\sqrt{2}$, and the two linearly independent solutions are:

$$F_1(u) = (1+u)^{\ell\sqrt{2}}, \qquad F_2(u) = (1+u)^{-\ell\sqrt{2}}.$$

Verified symbolically via SymPy: *both solutions give zero residual when substituted back into the ODE.*

Corollary 9 (Phase-Matching Condition — Mode $\ell = k$)**.** *Each Fourier mode $e^{i\ell v}$ carries angular momentum ℓ. The gauge connection (Corollary 2) contributes holonomy $2\pi k$ per cell. The relative phase mismatch per cell is:*

$$\delta\phi_\ell = 2\pi(\ell - k).$$

Resonant mode *($\ell = k$): $\delta\phi_k = 0$. The mode propagates coherently across cells with no phase drift.*

Non-resonant modes *($\ell \neq k$): Over N cells the accumulated phase is $2\pi N(\ell - k)$. For $N \gg 1$, these contributions destructively interfere and the mode decays.*

This identifies $\ell = k$ as a ***natural phase-matching condition****: the mode whose angular momentum matches the winding number is the one selected by coherence.*

Caveat. *This condition arises from phase alignment, not from a derived dynamical suppression law. No claim is made here that non-resonant modes are forbidden—only that they fail to propagate coherently across many cells. A rigorous decay estimate (e.g. bounding $\|T_\ell^N\|$ for $\ell \neq k$) would strengthen this to a derived selection rule, and is deferred to later work.*

Verified symbolically via SymPy.

Corollary 10 (Structural Correspondences: Wave Equation, Factorisation, and Normal Form)**.** *The Euler ODE from Corollary 8,*

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0,$$

admits three algebraic reductions. Each exhibits structural correspondence with a known equation of physics. No ansatz is used—only operator algebra and change of dependent variable. The correspondences are presented as ***structural analogies, not claimed equivalences****.*

1. Wave equation structure (Klein–Gordon analogy). *The Euler ODE arises from $\Box_{\mathcal{M}}\psi = 0$ after separation of variables. It is the full second-order wave equation on the manifold. This provides a Klein–Gordon-type structure, but relies on the* imported *Laplace–Beltrami operator (Corollary 8).*

2. Scalar first-order factorisation (Dirac analogy). *Define the Euler operator $D = (1+u)\, d/du$. Then:*

$$D^2F = (1+u)\frac{d}{du}\Big[(1+u)F'\Big] = (1+u)^2F'' + (1+u)F'.$$

So the Euler ODE is $D^2F - 2\ell^2F = 0$, which factors algebraically:

$$\big(D - \ell\sqrt{2}\big)\big(D + \ell\sqrt{2}\big)F = 0.$$

Each factor is a first-order equation:

$$(1+u)\, F' + \ell\sqrt{2}\, F = 0,$$
$$(1+u)\, F' - \ell\sqrt{2}\, F = 0.$$

Each reproduces one of the two indicial solutions:

$$F = (1+u)^{-\ell\sqrt{2}} \quad \textit{and} \quad F = (1+u)^{+\ell\sqrt{2}}$$

respectively. The factorisation is verified by checking that both residuals vanish.

Caveat. *This is a* scalar *first-order factorisation. It is structurally analogous to Dirac's procedure of factoring a second-order operator into first-order pieces, but it does not introduce spinors, Clifford algebra, or additional internal degrees of freedom. The analogy is algebraic, not physical, until further structure is derived.*

3. Schrödinger-type normal form. *To eliminate the first-derivative term from the Euler ODE algebraically, substitute $F = (1+u)^\alpha\, h(u)$ and choose α to kill the h' coefficient. The requirement is $2\alpha + 1 = 0$, giving $\alpha = -1/2$. With $F = (1+u)^{-1/2}\, h(u)$:*

$$(1+u)^2\, h'' + \big(\tfrac{1}{4} - 2\ell^2\big)\, h = 0.$$

No h' term remains. Dividing by $-2(1+u)^2$:

$$-\frac{1}{2}\, h'' + \frac{2\ell^2 - \frac{1}{4}}{2(1+u)^2}\, h = 0.$$

This is a Schrödinger-type normal form (Sturm–Liouville structure): $-\frac{1}{2}h'' + V_{\text{eff}}(u)\,h = 0$ with effective potential $V_{\text{eff}} = (2\ell^2 - 1/4)/(2(1+u)^2)$, which for $\ell \gg 1$ approaches the geodesic potential $\ell^2/(1+u)^2$ from Corollary 5.

Caveat. *This is a* time-independent *normal form. It is not yet a full time-dependent Schrödinger equation; the role of u as an evolution parameter is not formally established within the present axioms.*

4. Coordinate sensitivity: origin of i. *In the gauge-diagonalised cone frame, the ODE is purely real. In the* original (u, v) *frame, the inverse metric has $g^{uv} = -\pi k$, and the cross-term $g^{uv}\,\partial_u\partial_v\psi$ introduces imaginary contributions. On the mass shell ($\ell = k$), the coefficient of $f'(u)$ separates as:*

$$\textit{coeff}(f') = \underbrace{\frac{1}{2(1+u)}}_{\textit{real: geometric damping}} - \underbrace{2i\pi k^2}_{\textit{imaginary: constant}}.$$

The imaginary part $-2i\pi k^2$ is constant because $g^{uv} = -\pi k$ is constant, and it vanishes in the cone frame because the gauge diagonalisation removes g^{uv} entirely.

Caveat. *This imaginary structure is* coordinate-dependent*: it is present in the (u, v) frame and absent in the (u, v') frame. Its physical significance (if any) remains to be established. Both frames describe the same geometry; the i is a feature of the chart, not yet shown to be a feature of the physics.*

Summary. *Three levels of the same equation:*

Structure	***Operation***	***Result***	***Status***
Wave eqn	*Full Euler ODE*	$(1+u)^2F'' + (1+u)F' - 2\ell^2F = 0$	*Imported* $\Box_{\mathcal{M}}$
Factorisation	$(D-r)(D+r)F = 0$	*Two first-order equations*	*Scalar only*
Normal form	$F = (1+u)^{-1/2}h$	$-\frac{1}{2}h'' + V_{\text{eff}}\,h = 0$	*Time-indep.*

Verified symbolically via SymPy: *operator factorisation residuals zero; normal-form reduction confirmed; f' coefficient splits into real and imaginary parts as stated.*

4 Interpretation Boundary

The present work distinguishes between three categories of result:

Derived from axioms and standard differential geometry:

Metric · Connection · Holonomy · Christoffel symbols · Geodesics · Gaussian curvature · Cell action · Cell-independence · Fourier quantisation · Phase-matching condition

Imported from standard Riemannian geometry:

Laplace–Beltrami operator · Wave equation $\Box_{\mathcal{M}}\psi = 0$

Interpretive (structural correspondence, not equivalence):

Klein–Gordon analogy · Dirac-like factorisation · Schrödinger normal form · Phase-matching as "mass shell" · Imaginary coefficient as physical i

These correspondences are suggestive. Whether any of them can be promoted to physical equivalences requires: (a) a derived identification of u with a physical evolution parameter; (b) an observer or measurement framework; and (c) predictions that can be tested against experiment.